

Notes Risk Management Lab5 - CCR Exposures

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1 Extended Vasicek model

The extended Vasicek model (also known as the one-factor Hull–White model) for the short-term interest rate r_t at time t on the interval $(t, t + dt)$ is given by:

$$r_t = \varphi(t) + x_t,$$

where $\varphi(t)$ is a deterministic function of time, and x_t is an Ornstein–Uhlenbeck (OU) process

$$\begin{cases} dx_t &= -ax_t dt + \sigma dW_t \\ x_{t_0} &= x_0 \end{cases}$$

Here, $a > 0$ is the (constant) mean-reversion speed of the process and σ the (constant) volatility of the short rate.

While the parameters a and σ must be calibrated from market instruments (typically swaptions or caps/floors), the function $\varphi(t)$ is chosen so as to exactly fit the term structure of spot interest rates.

Question 1.1. *Why is it necessary to perfectly fit market data?*

Solution of the Ornstein-Uhlenbeck (OU) process

$$x_t = x_0 e^{-a(t-t_0)} + \sigma \int_{t_0}^t e^{-a(t-u)} dW_u.$$

Distribution of x_t Since x_t is a linear functional of a Brownian motion, is Gaussian with known mean and variance:

$$\mathbb{E}[x_t] = x_0 e^{-at}$$

$$Var(x_t) = \frac{\sigma^2}{2a} (1 - e^{-2at})$$

Forward discount factor in the OU framework In this OU framework, at time t the forward discount factor between maturities T_1 and T_2 satisfies

$$\frac{B(t, T_1, T_2)}{B(t_0, T_1, T_2)} = \exp \left(-x_t \left(\frac{\sigma(t, T_2) - \sigma(t, T_1)}{\sigma} \right) - 1/2 \int_{t_0}^t [\sigma^2(u, T_2) - \sigma^2(u, T_1)] du \right) ,$$

where

$$\sigma(u, \tau) = \sigma \frac{1 - e^{-a(\tau-u)}}{a}$$

Question 1.2. Compute the discount factor $B(t, \tau)$.

Question 1.3. Compute $\mathbb{E}_0[B(t, \tau)]$.

Question 1.4. How is the model calibrated?

Suggestions for numerical efficiency (affine form):

$$B(s, \tau) = A(s, \tau) e^{-x_s C(s, \tau)}$$

where

$$A(s, \tau) = \frac{B(t_0, \tau)}{B(t_0, s)} e^{-\frac{1}{2} \int_{t_0}^s (\sigma(u, \tau)^2 - \sigma(u, s)^2) du}$$

and

$$C(s, \tau) = \frac{1 - e^{a(\tau-s)}}{a}$$

Simulate a portfolio value

1. Write the portfolio value at time t in terms of x_t . Question: how do you do it for a swap? .
2. Simulate x_t on the selected grid. Question: what is the law of $x_{t+\Delta t}$ given x_t ?
3. Evaluate the portfolio price in the different MC scenarios.

2 Counterparty Credit Risk (CCR)

Counterparty Credit Risk (CCR) is the risk that one of the parties involved in a financial contract will be unable to meet its obligations before the maturity of the contract, thereby causing a loss to the other party.

This risk is primarily associated with **over-the-counter (OTC) markets**, where contracts are privately negotiated and exposure is borne directly by the counterparties. By contrast, CCR is far less relevant for **exchange-traded derivatives**, which rely on standardized contracts, daily mark-to-market procedures, margin requirements, and when there is the interposition of a **central counterparty (CCP)**, all of which significantly mitigate counterparty risk.

CCR is typical of contracts with deferred cash flows and uncertain values that depends on market conditions, such as **derivatives** (interest rate swaps, currency swaps, options, commodity swaps...) traded OTC. At inception, these contracts usually have a value close zero or very low, but it evolves over time. The risk materializes when a contract has a positive value for one party and the counterparty defaults.

CCR differs from **traditional credit risk** in several aspect.

Regarding traditional credit risk:

- it is typical of **loans and bonds**;
- exposure **known ex ante**;
- exposures evolves in a relatively **predictable** manner;
- it is a **unilateral** relationship: the creditor is exposed to the debtor and not vice versa;
- it arises at the moment the **principal is disbursed**.

By contrast, counterparty credit risk:

- is typical of **derivatives and financial contracts**
- the exposure is **not known at inception**;
- exposure is **time-varying** as it depends on **market factors**;
- it is a **bilateral** relationship: each party may alternate between creditor and debtor;
- it may emerge **long after contract inception**.

Historically, modern Counterparty Credit Risk emerged with the development of contemporary financial markets. While counterparty risk was minimal before the twentieth century due to simple instruments and immediate settlement, it became significant from the 1970s onward after the collapse of the Bretton Woods system increased market volatility. The rapid expansion of OTC derivatives, initially for hedging and later as a core banking activity, brought CCR to the forefront, though it long remained underestimated. Its systemic nature was fully revealed during the 2007–2008 global financial crisis, particularly following the failure of Lehman Brothers, leading to post-crisis reforms such as Basel III, the introduction of CVA capital charges, greater use of CCPs, and stricter collateral requirements to enhance financial stability.

2.1 Risk metrics: EPE, ENE, PFE, Peak-PFE

Consider party B and counterparty C. The stochastic process MtM_t : it denotes the net mark-to-market value at time t of a portfolio of instruments exchanged between B and C, belonging to the same netting set (see following paragraph), measured from the

perspective of party B .

If $MtM_t > 0$, the portfolio represents an asset for party B and a liability for counterparty C . In the event that counterparty C defaults at time t , party B is exposed to a potential loss equal to MtM_t , since C may fail to repay the amount owed in full.

Conversely, if $MtM_t < 0$, the portfolio represents a liability for party B and an asset for counterparty C . In this case, upon default of counterparty C , standard close-out agreement requires party B to settle the amount by paying $|MtM_t|$ to C . As a result, party B does not incur any credit loss.

The same reasoning applies symmetrically when the situation is viewed from the perspective of party C , in the case of party B defaulting.

When collateral (see following paragraphs) is posted, the potential loss reduced: the relevant exposure is no longer MtM_t , but rather $Exposure_t = MtM_t - Collateral_t$.

In light of these observations, and adopting the contemporary industry definitions (noting that Basel regulations originally introduced slightly different terminology), we define the following quantities from the perspective of party B : alled negative expected exposure).

- **Expected positive exposure** $EPE(t) = E_0[Exposure_t^+]$, the average amount that party B can lose in the event that counterparty C defaults at time t (when first introduced, it was called simply expected exposure).
- **Expected negative exposure** $ENE(t) = E_0[Exposure_t^-]$, the average amount that counterparty C can lose in the event that counterparty B defaults at time t (it is sometimes called negative expected exposure).

Question 2.1. *Given these definitions, why netting of the positions is good?*

These two quantities constitute the cornerstones of **Credit Valuation Adjustment CVA** and **Debit Valuation Adjustment DVA**.

Under a set of strong simplification assumptions (constant LGD , independence of defaults and exposures, unilateral default) we can briefly introduce these two concepts with the following approximated definitions:

- $CVA_0 = LGD_C \int_0^T EPE_0(t) B_0(t) S_C(0, t) dPD_C(t)$
- $DVA_0 = LGD_B \int_0^T ENE_0(t) B_0(t) S_B(0, t) dPD_B(t)$

$B_0(t)$ is the risk-free discount factor, $S_C(0, t)$ is the survival probability of counterparty C up to time t , and $dPD_C(t)$ represents the incremental default probability of C at time t , given that C survived up to t , $S_B(0, t)$ is the survival probability of party B up to time t , and $dPD_B(t)$ represents the incremental default probability of B at time t , given that B survived up to t .

Question 2.2. *What needs to be changed if we suppose that both default of B and C can happen?*

Question 2.3. *Why shouldn't exposures and probability of default be considered independent?*

The credit-risk-adjusted mark-to-market value of the portfolio at time t is given by, from B perspective:

$$MtM_t = MtM_t^{risk-free} - (CVA_t + DVA_t)$$

Potential Future Exposure (PFE) represents a severe but plausible exposure scenario at a given confidence level over a specified future time horizon. It can be interpreted as the VaR analogue for credit exposures.

$$PFE_\alpha(t) = \text{Quantile}_\alpha(\text{Exposure}^+(t)).$$

It is a measure of counterparty credit risk that represents a high-quantile estimate of future exposure.

It is commonly used in credit limit setting, stress testing, and scenario analysis, and it is one of the key Basel building blocks for the calculation of Exposure at Default (EAD).

The **peak Potential Future Exposure** (peak PFE) is defined as the maximum value of the potential future exposure over a given time horizon,

$$Peak - PFE_\alpha = \sup_{t \in [0, T]} PFE_\alpha(t).$$

It provides a conservative, time-agnostic measure of the maximum exposure that may arise with a given confidence level over the life of a portfolio or transaction. It is primarily used in the context of **credit limit setting and monitoring**.

Since credit limits are often expressed as a single scalar value rather than a full time-dependent exposure profile, institutions compare the peak PFE to the approved limit to ensure that exposures remain within the defined risk appetite throughout the entire time horizon.

As such, peak PFE is well suited as a prudential risk control metric and for high-level risk reporting.

Counterparty risk can be **mitigated** through contracts and more complex mechanisms, but it can never be fully eliminated. Risk mitigants always leave residual counterparty risk, which is often harder to measure than the original risk. Importantly, mitigants usually **transform** counterparty risk into other types of financial risk rather than removing it altogether (e.g. liquidity risk, legal risk, operational risk, or market risk).

Break Clauses, **Netting** and **Collateral** can be considered in this framework as risk mitigant.

2.2 Break Clauses

Break clauses are contractual provisions that allow one (unilateral) or both (bilateral) parties to terminate or restructure a financial transaction either mandatorily, optionally, or upon the occurrence of a defined event, most commonly a deterioration in credit quality such as a rating downgrade.

They are widely used in derivatives and other long-dated financial contracts as tools to manage counterparty credit risk.

Historically, before the global financial crisis, break clauses were mainly demanded by banks when trading with weaker or uncollateralised counterparties. After the GFC, concerns about bank creditworthiness led asset managers, pension funds, and other clients to increasingly require break clauses linked to banks' own credit ratings.

Mandatory breaks are sometimes used to bypass **internal credit limits**, often with an informal expectation that the trade will be re-executed at the break date.

Exercising a break clause does not always lead to immediate termination. Alternatives may include providing a replacement counterparty, and in case of termination, the trade is usually settled at its prevailing mark-to-market (MTM) value, often determined via dealer quotations, which is generally simpler and less contentious than close-out valuation used in default scenarios. From a risk and regulatory perspective, break clauses differ significantly in effectiveness:

- **Counterparty risk reduction:** all break clauses are viewed as beneficial, though optional and rating-based breaks are less reliable because their exercise is discretionary or dependent on lagging indicators.
- **Liquidity coverage ratio risk:** rating-based triggers are particularly costly because banks must pre-fund potential liquidity needs arising from them.
- **Capital relief:** it is relief is generally achievable only for mandatory breaks. Optional breaks rarely qualify unless they are exercised consistently, while Basel rules explicitly disallow capital relief from rating-based triggers.

On the other hand, break clauses come at a cost:

- **Moral hazard:** sales teams may push for deal execution by relying on the presence of a break clause to obtain internal credit approval and to offer longer maturities to clients. However, once the trade is in place, sales may oppose the exercise of the break from the bank's side in order to preserve the client relationship, resulting in increased xVA position and higher RWA consumption;
- **Reliance on credit rating:** they are typically slow-moving indicators of credit quality, with downgrades often occurring only once a counterparty's financial distress has already become severe.
- **Cliff-edge effects:** the widespread use of similar rating-based triggers across counterparties can generate cliff-edge effects, whereby relatively small events, simultaneously activate break clauses, collateral calls, or termination rights. This can lead to sudden and system-wide liquidity stress.
- **Model risk:** break clauses are hard to model and hedge. Rating transition probabilities cannot be inferred from market prices and rely on limited historical data, making valuation uncertain. Mandatory breaks are the main exception, as they can be modeled deterministically.

2.3 Netting

A **close-out netting agreement**, such as the ISDA Master Agreement, provides the legal framework that allows assets and liabilities arising from derivative transactions (of any maturity, whether in- or out-of-the-money) to be netted and collapsed so that only a single net amount is owed when a counterparty defaults.

If the surviving party owes money, then it makes this payment; if it is owed money, then it makes a bankruptcy claim for that amount.

Under this legal agreement between two counterparties, derivative trades can be grouped into a **netting set**, which represents the level at which counterparty risk and related **risk metrics** are calculated.

Two counterparties may have multiple netting sets, for example, to reflect different jurisdictions, legal entities, asset classes, or regulatory requirements. In some cases, individual trades may also be treated as separate netting sets.

In rare cases where netting is not permitted or its enforceability is uncertain, each trade is treated as a standalone netting set, resulting in higher counterparty exposure.

Consider two parties, *Bank B* and *Counterparty C*, trading under an ISDA Master Agreement that includes a valid close-out netting clause.

From perspective of party *B* towards counterparty *C* the set of trades in place is composed by:

- Interest rate swap: +10 (*B* is owed 10 by *C*);
- FX forward: -6 (*B* owes 6 to *C*);
- Equity option: +4 (*B* is owed 4 by *C*).

Without netting:

- Gross amount owed to *B*: $10 + 4 = 14$;
- Gross amount owed by *B*: 6.

Multiple obligations remain outstanding between the two counterparties.

With netting, all positions are offset:

$$\text{Net exposure} = 10 - 6 + 4 = 8.$$

Therefore only a single net amount of 8 is owed by *C* to *B*.

The benefits of a netting agreement are:

- a reduction in exposures and, therefore, in counterparty credit risk;
- lower collateral posting requirements;
- a reduction in the number of cash flows exchanged;
- a simplification of the close-out process in the event of default;

- lower regulatory capital requirements, in terms of exposure at default (EAD) and risk-weighted assets (RWA).

On the other hand, netting may create or amplify the following risks:

- **Legal risk:** in the presence of different jurisdictions or inadequate legal frameworks, close-out netting may not be enforceable and the expected benefits may fail to materialize;
- **Systemic risk:** netting increases interconnectedness among market participants, so the default of a single counterparty can transmit losses and liquidity stress across the financial system, amplifying contagion during periods of crisis;
- **Concentration risk:** by aggregating many transactions into one net exposure, netting concentrates risk toward a small number of counterparties, making losses larger and more severe if one of them fails.

Question 2.4. *Does netting improve also PFE?*

2.4 Collateral

The ISDA Master Agreement may be supplemented by a Credit Support Annex (CSA), which governs the exchange of collateral.

Collateral consists of cash or assets set aside, like money kept "in a safe", and is exchanged between counterparties as a safeguard to cover exposures and reduce potential losses in the event of default.

By limiting unsecured exposures arising from changes in the mark-to-market value of contracts, collateral mitigates counterparty credit risk and supports overall market stability within modern regulatory frameworks.

In doing so, it converts counterparty risk into:

- **Liquidity risk:** the ability to source sufficient collateral, particularly during periods of market stress;
- **Market risk:** fluctuations in the value of posted collateral and the presence of wrong-way risk, especially if collateral is non-cash asset;
- **Operational risk:** risks related to margining processes, valuation disputes, and settlement failures.

The **Credit Support Annex (CSA)** is the legal document governing the collateralization process. It defines the eligible collateral and sets the rules for their valuation and exchange.

The CSA specifies:

- **eligible assets** that may be posted as collateral, such as cash or specified securities, usually government, supranational or investment-grade bonds,

- **haircuts**, a percentage discount applied to the market value of the collateral in order to account for price, liquidity, and foreign exchange (FX) risks, when the posted assets are non-cash or denominated in a different currency.

Then, the total collateral is composed of two components **Independent Amount** (IA) and **Variation Margin** (VM):

- **independent amount** (IA), is a portion of collateral always posted independently of the current mark-to-market of the netting set. It is typically fixed over time and is returned only upon termination of the transaction, provided that no default has occurred. Initial margin is intended to provide extra protection buffer between two margin calls and to cover potential losses and costs arising during the close-out period following a default. Historically, IA has been uncommon in bilateral OTC derivative transactions, while the almost equivalent initial margin (IM) has long been standard practice for exchange-traded derivatives and transactions cleared through central counterparties (CCPs).
- **variation margin** (VM), reflects changes in the mark-to-market *MtM* value of the positions and is adjusted dynamically over time. It serves to cover the variability of the portfolio of transactions in question and has long been a common feature of OTC derivative collateral arrangements.

The VM exchange conditions are governed by CSA parameters such as:

- a **margining frequency**, most commonly daily between large institutions - though weekly and monthly frequencies are standard - which determines how often exposures are recalculated and collateral is adjusted, through a **margin call**, the formal request to post additional or return excess collateral;
- a **threshold**, representing the level of exposure that may remain unsecured. If $MtM < Threshold$, there is no need to post VM, while if $MtM > Threshold$, only the excess exposure above the threshold is subject to a margin call;
- a **minimum transfer amount** (MTA), which is the minimum amount below which no margin call is triggered, to prevent operationally inefficient small transfers, in particular of non-cash margin;
- a **cap** which may limit the maximum amount of collateral that can be demanded, balancing risk mitigation with liquidity management and operational efficiency.

In the CSA other details are defined, such as:

- the **valuation agent**, **valuation date** and **valuation time**, *i.e.* the agent responsible for the evaluation of *MtM* on a specific date and time, to avoid litigation;
- **rounding rules** to apply to the margin call amount to simplify the transfers;
- the **cut-off time** and **settlement period**, the time of the day before which the margin call must be notified and the period of time allowed to transfer the collateral requested;

- the **interest** rate on cash collateral, that has to be remunerated (e.g. ESTR, Fed Funds...);
- some **rating triggers**, specifying that under a certain rating downgrade, the collateral rules might change.

Each parameter may take different values for each counterparty involved. Therefore, we define IA_B , IA_C , THR_B , THR_C , MTA_B , MTA_C , Cap_B , Cap_C , with B and C denoting the two counterparties.

- IA_B deposited by B , for the benefit of C ;
- THR_B the threshold up to which B is not required to post VM to C ;
- MTA_B minimum amount that B could be asked to transfer to C in a margin call;
- Cap_B maximum amount of total collateral that B might be required to post to C .

From B perspective, at each margin call the target VM to be posted (if positive B receives collateral, if negative B pays collateral) is

$$VM^* = \begin{cases} \max(MtM - THR_C, 0) & MtM \geq 0 \\ \min(MtM + THR_B, 0) & MtM < 0 \end{cases}$$

Then, the theoretical margin call, with VM currently posted collateral

$$\Delta VM = VM^* - VM_{t-1}$$

$$\text{Margin Call}_t = \begin{cases} \Delta VM_t, & \text{if } |\Delta VM_t| \geq MTA_C \wedge \Delta VM_t \geq 0 \\ \Delta VM_t, & \text{if } |\Delta VM_t| \geq MTA_B \wedge \Delta VM_t < 0 \\ 0, & \text{otherwise} \end{cases}$$

Then

$$VM_t = VM_{t-1} + \text{Margin Call}_t$$

and in case there are some cap conditions the effectively posted VM is

$$V\bar{M}_t = \begin{cases} \min(VM_t, Cap_C) & \text{if } VM_t \geq 0 \\ \max(VM_t, -Cap_B) & \text{if } VM_t < 0 \end{cases}$$

Here we are ignoring the "time evolution" of the posted collateral, *i.e.* the change in value of the asset posted as collateral between t_{i-1} and t_i , we are ignoring the haircuts and any rounding rule. We also suppose that IM is always posted and independent of VM .

Question 2.5. *From B perspective, which values for IA , THR , MTA , Cap would be preferred, for the counterparty and for itself?*